HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **analog, digital, and power electronics** circuitry for servo-drive products, meeting stringent performance and reliability targets.
- Developed **VHDL** code for FPGA implementation, ensuring efficient and high-speed digital logic operations for embedded motor control.
- Managed the **hard real-time embedded system** development lifecycle, adhering to project schedules and delivering functional firmware for complex control solutions.
- Collaborated with multi-site teams on hardware design reviews and business reviews, contributing to architectural decisions and product strategy.
- Integrated and tested **electronic devices** including A/D, D/A converters, and optical encoders with FPGAs for precise motor feedback and control.
- Contributed to product design for manufacture (DFM) and design for test (DFT) processes, ensuring manufacturability and efficient testing of electronic motor control solutions.
- Supported on-site customer implementations and troubleshooting for electronic control systems, addressing technical challenges in diverse operational environments.

SKILLS

Hardware Design – Analog Electronics, Digital Electronics, Power Electronics, Servo-Drive Design, DFM, DFT

Embedded Systems – Hard Real-Time Embedded Systems, FPGA, VHDL, Embedded C/C++

Motor Control – Permanent Magnet Synchronous Motors (PMSM), Motor Control Principles, Power Conversion Topologies

Control & Integration – A/D Converters, D/A Converters, Optical Encoders, Thermal Management

Development Process – Requirements Definition, Architectural Design, Integration, Testing, Design Reviews, Agile Methodologies

Support & Operations – On-site Customer Support, Troubleshooting, Regulatory Compliance

PROJECTS

High-Reliability Servo-Drive Control Unit (HCL Technologies)

- Developed the hardware architecture for a new servo-drive control unit, focusing on power electronics and digital signal processing to meet demanding reliability specifications.
- Implemented VHDL logic for FPGA-based motor control algorithms, optimizing for real-time performance and integration with analog feedback sensors.
- Conducted thorough design reviews and simulation cycles for analog and digital components, ensuring compliance with internal performance benchmarks and external regulatory standards.

Embedded System for Motor Performance Monitoring (MSc Project)

- Designed an embedded system using Python and C for real-time data acquisition from motor sensors (e.g., encoders, temperature probes) during MSc coursework.
- Developed data processing routines to analyze motor performance metrics, identifying anomalies and potential failure points under various load conditions.
- Integrated a basic reporting module to visualize key performance indicators, demonstrating principles of operational data analysis for electronic systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)